KOSZUL ORTHOGONALITY FOR OBJECTS SUPPORTED AWAY FROM AN OPEN OF AN AFFINE SCHEME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Koszul complexes fit into distinguished triangles under multiplication of the exponents, by the More on Algebra Koszul multiplicativity lemma. invoked at proof lines 4097-4100. Transfers the Hom vanishing from the Koszul complex K to all its powers K_e.
- For an object supported on the closed complement of U, its cohomology classes on X are represented by maps from the powers of the Koszul complex, by the represent-cohomology-class lemma. invoked at proof lines 4101-4105. Converts the Hom vanishing into the vanishing of the cohomology of N, hence N = 0.
- An object of D_QCoh on an affine scheme is determined by the complex of modules of its global derived sections. invoked at proof lines 4106-4107. Identifies the vanishing of R Gamma with the vanishing of the object.

Source reference. Original lemma 062C: sources/stacks-chow/tags.txt, source lines 7791–7791. Cited at proof lines 4100-4101 to pass from K to the powers K_e; the Hom vanishing transfer is inside the counted span.

Source reference. Original lemma 08E3: sources/stacks-perfect/pinned-original-source.tex, source lines 2374–2374. Cited at proof lines 4101-4105 to convert the Hom vanishing into the vanishing of R Gamma; the conclusion is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 4106-4107 to conclude E = 0; the equivalence is inside the counted span.

Source reference. Original remark 08CZ: sources/stacks-chow/tags.txt, source lines 10619–10619. Cited at statement line 4071 as the standing setup; the orthogonality argument is inside the counted span.

Lemma 0.2. *In Situation 08CZ denote $j : U \to X$ the open immersion and let K be the perfect object of $D(\mathcal{O}_X)$ corresponding to the Koszul complex on $f_1, \ldots, f_r$ over A. For $E \in D_{QCoh}(\mathcal{O}_X)$ the following are equivalent*

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(1) $E = Rj_*(E|_U)$, *and*
(2) $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[n], E) = 0$ *for all* $n \in \mathbf{Z}$.

Proof. Choose a distinguished triangle $E \to Rj_*(E|_U) \to N \to E[1]$. Observe that

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K[n], Rj_*(E|_U)) = \mathrm{Hom}_{D(\mathcal{O}_U)}(K|_U[n], E) = 0$$

for all n as $K|_U = 0$. Thus it suffices to prove the result for N. In other words, we may assume that E restricts to zero on U. Observe that there are distinguished triangles

$$K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i'+e_i''}, \ldots, f_r^{e_r}) \to K^\bullet(f_1^{e_1}, \ldots, f_i^{e_i''}, \ldots, f_r^{e_r}) \to \ldots$$

of Koszul complexes, see More on Algebra, Lemma 062C. Hence if $\mathrm{Hom}_{D(\mathcal{O}_X)}(K[n], E) = 0$ for all $n \in \mathbf{Z}$ then the same thing is true for the K replaced by K_e as in Lemma 08E3. Thus our lemma follows immediately from that one and the fact that E is determined by the complex of A-modules $R\Gamma(X, E)$, see Lemma 06Z0. □